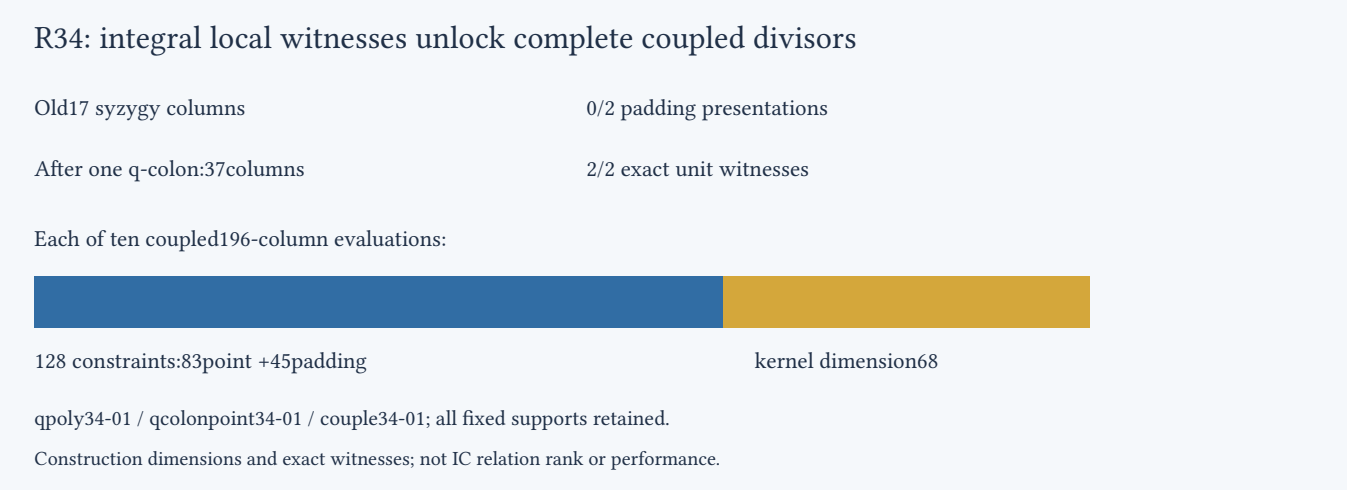
Koblitz cover transfers — R34

Observations and scoped proof consequences; no IC performance claim

The selected ordinary divisor construction now has exact source-bound local presentations at all ten degree83 supports and both degree5 padding points. The fixed degree83 points pass27product witnesses after682coefficient-domain checks. The same quotient basis is inadmissible at the padding points, where 566denominator entries vanish. This is a basis-domain exclusion, not a singularity result.



Primitive normalization does not repair the padding family: all17scales are1. Direct polynomial evaluation modulo the actual minimal q=u5+u3+1 verifies all old syzygy values and all68products are zero at bothQ. One exact module colon step computes20new integral vectors. Their identities qG=SH and JG=0 are checked BEFORE specializing atq=0; q is not inverted at the points. The enlarged37column family gives148products and exact unit witnesses at bothQ, preserving the old zero family as a negative control.

The standard-etale presentation argument gives local dimension1, beyond the earlier tangent tests. Binding to proper points uses the retained source function-field degree32 bridge and native generic vdim32, as detailed in LOCAL_GEOMETRY.md. This is a written geometric application, not a Lean proof of global normalization or a polarized factor embedding.

Complete nine-jets then pass all five source equations, eight units, every coefficient-descent conjugate,68descended-function comparisons and corrupt controls. Their196section evaluation has45constraints and kernel151. Combining it with each completed83point evaluation gives rank128 and kernel68 in every case. Original columns, separate point/padding constraints and all pivots are recombined independently; a perturbed kernel is rejected. These are complete spaces L(2D0-(P+9Q)), with actual multiplicity9 at the proper pointQ.

Each individual source class is nonprincipal by the earlier L(D0-P)=0 and the proper RR bridge. The signed projected class may still cancel, and no intended order11factor image is evaluated. Actual Picard operations remain. The next scaling lead is a closed-point evaluation algebra for154Q (degree770) to represent L3D0 and the proposed708dimensional L6D0; not yet constructed.

Two new Lean cancellation lemmas compile and pass the pinned transitive axiom audit,0failing0trusted. They explain integral cancellation before specialization; native arrays and proper geometry are not formalized. References: Stacks29.35.14/35.23.31; Singular syz,cleardenom,quotient; Khuri-Makdisi math/0409209v3 RepresentationB. Native4GiB/1200s per invocation, Lean4.34.1/mathlibd13f23b,-j1-M4096/1200s. No independent scientific replay, full quotient saturation, polarized factor embedding or unknown-scalar recovery.

Complete retained attempt inventory

An accepted diagnostic can record an excluded domain or a nonunit family. Native exit0 and a trailing marker do not override Singular errors. The first degree83consumer is rejected for the reserved identifier coeffs; the corrected fresh invocation uses identical points and mathematical conditions.

- couple34-01: accepted
- jets34-01: accepted
- lean34-01: accepted
- prefix34-check5-01: accepted
- prefix34-check83-01: rejected
- prefix34-check83-02: accepted
- prefix34-function01: accepted
- prefix34-verify01: accepted
- primitive34-01: accepted
- qcolon34-01: accepted
- qcolonpoint34-01: accepted
- qpoly34-01: accepted